Week 8 Technical Report: Explainable AI

Customer Churn Prediction Using Random Forest and SHAP

1. Introduction

This project investigates how and why a machine learning model predicts customer churn. A Random Forest Classifier is trained on the IBM Telco Customer Churn dataset. Permutation importance and SHAP are used to explain the model's behavior.

2. Objectives

- Train and evaluate a machine learning model.
- Identify the ten most important features.
- Generate and interpret SHAP visualizations.
- Analyze five individual predictions.
- Compare permutation importance and SHAP.
- Investigate potential bias and model limitations.
- Provide practical business recommendations.

3. Dataset and Model

The dataset contains telecommunications customer information, including contract type, tenure, services, payment method, and charges. The target variable is Churn, where 1 represents a customer who churned and 0 represents a customer who stayed.

The selected model is a Random Forest Classifier. The data is divided into training and test sets, and the model is evaluated on the held-out test data.

4. Model Evaluation

The model is evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and a confusion matrix. These metrics describe different aspects of predictive performance.

The actual evaluation scores are generated by the executed notebook and should be reported in this section.

5. Global Feature Importance

Permutation importance identifies features whose shuffling reduces the model's performance. The top ten features and their importance values are obtained from the test-set analysis.

6. SHAP Analysis

SHAP explains how features contribute to model outputs. The summary plot shows feature contributions across customers, the feature-importance plot ranks features by average absolute contribution, and the waterfall plot explains an individual prediction.

7. Individual Prediction Analysis

Five test-set customers are examined using their input values, actual outcomes, predicted outcomes, churn probabilities, and the strongest positive and negative SHAP contributions.

8. Comparison of Explanation Methods

Permutation importance measures the effect of feature shuffling on model performance. SHAP measures feature contributions to predictions. Differences between their rankings can arise because the methods measure different quantities and handle feature interactions and correlated inputs differently.

9. Potential Bias and Limitations

The analysis examines class distribution, model errors, and performance across selected customer groups. Differences between groups should be investigated, but they do not automatically prove unfairness.

Feature importance and SHAP describe model behavior rather than proving causal relationships. The model may also perform differently on new customers or future data.

10. Conclusion

This project demonstrates how a trained machine learning model can be evaluated and interpreted using permutation importance and SHAP. The findings provide evidence about the model's behavior on the selected dataset and help identify areas for further investigation.